Do Myers-Briggs Personality Dimensions Align with Neural Engagement Patterns During Formal Presentations?

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1. ABSTRACT

In classrooms and boardrooms alike, the moment of presentation often reveals unexpected contrasts. Individuals who appear socially confident in everyday interaction may display visible tension when addressing an audience, while others who are perceived as reserved may speak with measured composure. Such observations complicate widely held assumptions about personality and confidence. They invite a deeper question: do personality classifications meaningfully align with measurable neural engagement patterns during formal presentations?

The Myers-Briggs Type Indicator (MBTI) personality framework categorises individuals across four dimensions: introversion-extraversion, sensing-intuition, thinking-feeling, and judging-perceiving. These classifications are frequently used to interpret communication style and performance in educational and professional settings. However, such interpretations are typically derived from self-reported psychometric assessments rather than observable physiological evidence. This project, therefore, investigates whether MBTI personality dimensions correspond with measurable neural engagement patterns during formal presentation environments.

The study adopts a data-driven approach utilising anonymised physiological data collected from undergraduate students at Middlesex University through the Muse 2 wearable Electroencephalography (EEG) device. Individual-level brain bandpower signals will be processed and aggregated to identify physiological response patterns using unsupervised clustering techniques, enabling speakers with similar neural profiles to be grouped without prior personality labelling. These clusters will then be compared against each Myers-Briggs Type Indicator (MBTI) personality dimension (introversion–extraversion, sensing–intuition, thinking–feeling, and judging–perceiving) to assess the extent to which self-reported personality orientations align with observed neural engagement patterns. Emotional intelligence measures are incorporated as a complementary interpretive framework to support the analysis of stress regulation and attentional stability during presentations.

By focusing on measurable neural indicators such as alpha and beta band activity, the study moves beyond subjective interpretations of confidence and communication ability. It reframes presentation performance as a quantifiable physiological phenomenon that can be analysed using wearable sensor data and visualised through analytical dashboards. The study is expected to provide empirical insight into whether personality dimensions exhibit statistically meaningful associations with neural engagement clusters, or whether physiological response patterns operate independently of traditional personality classifications. In doing so, it contributes to a more evidence-based understanding of performance under evaluation.

2. DESCRIPTION OF THE PROJECT

2.1. Problem Definition

Personality classifications derived from frameworks such as the Myers-Briggs Type Indicator (MBTI) are widely used to explain differences in confidence and performance during formal presentations in educational and professional contexts. The MBTI categorises individuals across four dimensions: introversion–extraversion, sensing–intuition, thinking–feeling, and judging–perceiving. These classifications are commonly used to interpret communication style and behavioural tendencies in evaluative environments.

However, empirical research has reported weak relationships between MBTI dimensions and observed performance outcomes and highlighted that the instrument does not measure neuroticism, the personality trait most closely associated with emotional instability and stress reactivity (Fleenor, 1997).

Table 1.
Correlations Between MBTI Dimensions and Leaderless Group Discussion Performance Ratings (Competitive Exercise, N = 5,793).

Correlations between MBTI Scores and Leaderless Group Discussion Ratings

Competitive Exercise (N = 5,793)

	MBTI Scores			
LGD Ratings	EI	SN	TF	JP
Activity Level	-.15	.16	.00	.04
Led Discussion	-.15	.17	.00	.03
Influenced Others	-.12	.16	.00	.03
Problem Analysis	-.11	.16	.00	.03
Task Orientation	-.13	.16	.00	.02
Motivated Others	-.12	.15	.02	.03
Interpersonal Skills	-.06	.10	.04	.01
Verbal Effectiveness	-.09	.16	.01	.06
Overall Rating	-.13	.17	.01	.03

Source: Adapted from (Fleenor, 1997).

In formal presentation contexts, where attentional demand and emotional regulation play central roles, this limitation reduces the explanatory power of personality-based assumptions. Evaluative judgments about presentation performance are therefore frequently formed without reference to measurable neural indicators of attentional engagement or emotional regulation. This creates a gap

between assumed personality traits and observable engagement patterns during performance, particularly in contexts where cognitive focus and regulatory capacity may vary independently of self-reported personality dimensions.

2.2. Why It Is Important

This study is important because it reframes confidence in formal presentations as a measurable behavioural and neural response rather than a fixed personality trait. In educational and professional settings, confidence is frequently inferred from psychometric classifications, despite evidence that such tools do not reliably predict observed performance under pressure. Empirical research indicates that the Myers–Briggs Type Indicator (MBTI) exhibits weak relationships with behavioural performance ratings and does not capture neuroticism, a dimension closely associated with stress reactivity (Fleenor, 1997). In information technology education, communication and behavioural competencies have been shown to significantly influence outcomes beyond technical expertise alone (Rodríguez Montequín et al., 2012).

More fundamentally, this study addresses a measurement gap in which high-stakes evaluative decisions are often informed by self-reported personality data rather than observable physiological evidence. By grounding the analysis in neural bandpower signals collected through the Muse 2 wearable electroencephalography (EEG) device, the study introduces greater empirical precision and reduces reliance on assumption-based interpretations of personality dimensions. By examining attentional engagement patterns and stress-related neural indicators, the project replaces inference-based evaluations of confidence with data-driven behavioural analysis, offering both theoretical insight into personality-performance relationships and practical technological contributions.

2.3. Who Will Benefit

This research is relevant to students, educators, organisations, and the broader professional environments in which presentation performance influences evaluation and decision-making.

Students may benefit from more evidence-informed approaches to presentation assessment, particularly in high-stakes academic contexts where performance outcomes contribute to degree classification and future opportunities. By examining neural engagement patterns during presentations, the study encourages evaluation practices grounded in observable physiological indicators rather than relying solely on inferred personality classifications.

Educators and assessors may benefit from improved fairness and consistency in performance evaluation through data-driven insight into attentional stability and stress-related neural activity. In organisational contexts, effective presentations shape decision-making, stakeholder confidence, and communication outcomes (Swathi, 2015).

By reframing presentation performance as a measurable engagement process rather than a fixed personality characteristic, the study offers a basis for more nuanced training, feedback, and professional development strategies.

3. RATIONALE AND BACKGROUND

3.1. State of the Art

Research on communication performance in educational and organisational settings has developed along three largely parallel strands.

First, personality-based frameworks such as the Myers–Briggs Type Indicator (MBTI) are frequently used to categorise individuals according to preferences in perception and decision-making, with particular attention often given to the introversion-extraversion dimension as indicative of presentation style and confidence (Fleenor, 1997). Such frameworks remain widely applied in leadership development and performance discussions. However, empirical investigations have reported weak relationships between MBTI classifications and observed communication effectiveness, raising questions about the predictive validity of personality classifications in performance contexts (Fleenor, 1997). Additionally, the MBTI does not assess neuroticism, the trait most closely associated with stress reactivity, which is particularly relevant in high-pressure presentation settings.

Second, research in behavioural analytics has examined presentation environments through structured observation, visual analysis, and performance-rating systems. Engagement and interaction patterns have been shown to be systematically captured and visualised, reflecting a growing emphasis on measurable performance indicators rather than self-reported traits (Dafoulas et al., 2020). While these approaches provide objective insight into observable behaviour, they primarily capture external conduct and rarely incorporate direct physiological or neural measurement of attentional engagement during performance.

Third, studies in educational and organisational contexts emphasise the role of communication competence and emotional regulation in effective performance (Rodríguez Montequín et al., 2012; Swathi, 2015). Emotional intelligence research addresses stress management and interpersonal effectiveness, yet it is typically examined independently of personality typologies and real-time physiological indicators.

Personality typologies, behavioural measurement systems, and emotional regulation frameworks therefore continue to evolve as largely distinct lines of inquiry, with limited empirical integration across psychometric, behavioural, and physiological levels of analysis.

3.2. Existing Approaches

Current solutions for assessing and improving presentation performance operate across three main areas: psychometric assessment, behavioural analytics, and immersive technological intervention.

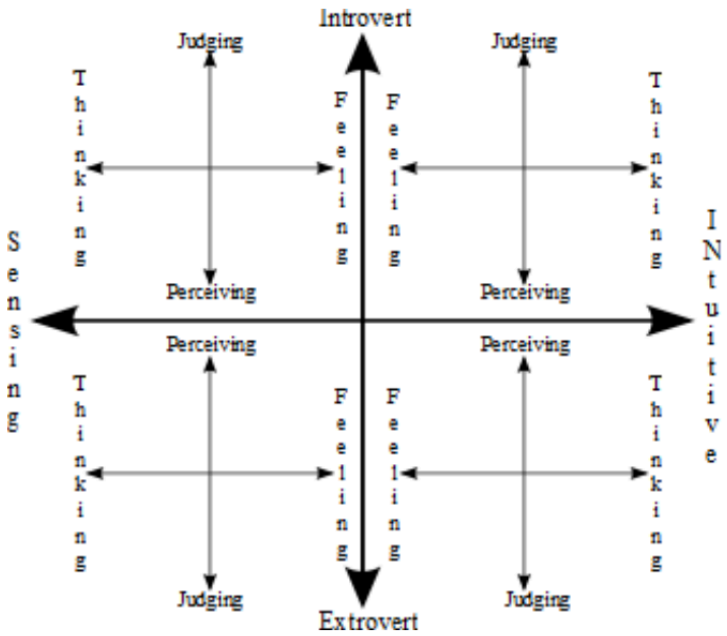

Figure 1. Structural representation of the Myers–Briggs Type Indicator (MBTI) four-dimension framework (Introversion–Extraversion, Sensing–Intuition, Thinking–Feeling, Judging–Perceiving). Source: Adapted from (Rodríguez Montequín et al., 2012).

Psychometric instruments such as the Myers-Briggs Type Indicator (MBTI), broader trait-based inventories, emotional intelligence assessments, and communication apprehension scales are commonly used to profile personality orientation, emotional regulation, and communication anxiety in performance settings. Weak associations between MBTI classifications and behavioural effectiveness have been reported, raising concerns regarding predictive validity in performance contexts (Fleenor, 1997).

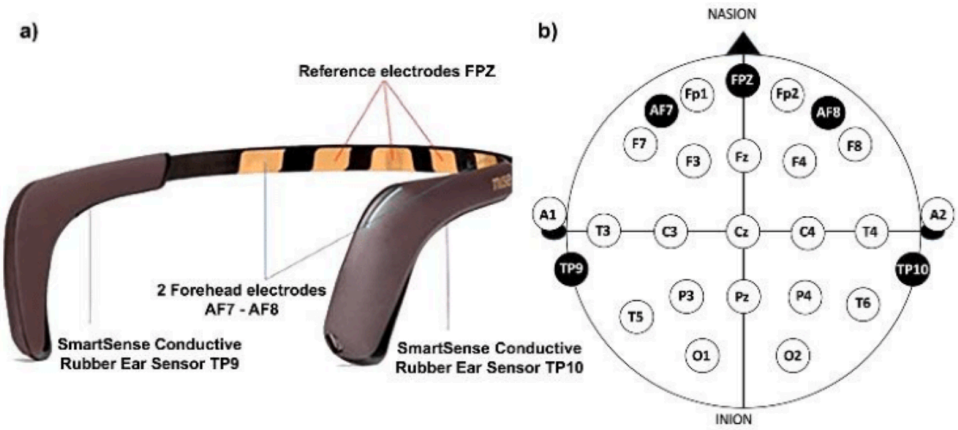

Figure 2. Electrode placement of the Muse 2 wearable Electroencephalography (EEG) device, illustrating frontal (AF7, AF8) and temporal reference (TP9, TP10) sensor positions within the international 10–20 system.
Source: Adapted from (InteraXon, 2022).

Alongside self-report profiling, behavioural and physiological systems generate objective performance data through structured sensor integration, transforming raw biological signals into

interpretable indicators. Engagement and interaction patterns have been captured through structured observation and visual analytics approaches (Dafoulas et al., 2020). Gaze-monitoring systems have quantified eye-contact distribution and attentional allocation during decision-making and performance tasks (Du et al., 2013).

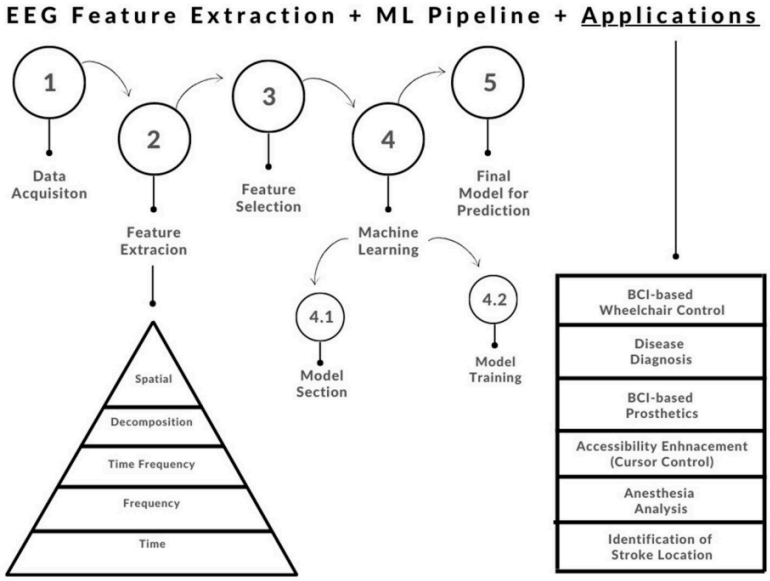

Figure 3. General EEG feature extraction and machine learning pipeline, illustrating stages from data acquisition to predictive modelling.
Source: Adapted from (Singh and Krishnan, 2023), based on (Subasi, 2019).

Technological training systems extend these capabilities into rehearsal environments. Virtual reality platforms simulating audience presence have demonstrated statistically significant improvements in clarity and observer-rated presentation performance (Bachmann et al., 2023). AI-based feedback systems further extract multimodal indicators such as vocal pacing, filler-word frequency, facial expressiveness, and speech clarity to provide structured performance evaluation.

While high-density clinical EEG systems provide greater spatial resolution, they are expensive, laboratory-bound, and intrusive. In contrast, the Muse 2 wearable EEG device offers portability, affordability, and ease of deployment in naturalistic classroom environments. Although it provides fewer channels than research-grade systems, its frontal electrode placement (AF7, AF8) makes it particularly suitable for measuring attentional engagement and cognitive activation during live presentation tasks. For this study, ecological validity and real-world feasibility are prioritised over high-density signal acquisition, making Muse 2 an appropriate and methodologically justified technological choice for this study.

3.3. Limitations of Existing Approaches

Despite significant methodological advancement, existing solutions remain analytically fragmented.

Psychometric instruments profile personality preference and emotional tendencies, yet rely primarily on self-reported data that does not capture real-time behavioural or physiological response. Behavioural and sensor-based systems generate objective metrics derived from observable behaviour or physiological signals, but typically operate independently of structured

personality frameworks. Training technologies simulate performance environments and provide automated feedback, yet rarely integrate personality orientation, emotional regulation capacity, and physiological response within a unified analytical model.

Recent AI-driven feedback systems further illustrate this fragmentation. For example, automated large language model-based feedback systems generate structured presentation evaluations at scale; however, evaluation findings have highlighted variability in feedback outputs and the continued need for human oversight (Gürtl et al., 2025). While such systems enhance scalability and automation, they do not incorporate validated personality constructs or physiological measurement, and the interpretability of algorithmic decision processes remains limited.

As a result, confidence in presentation contexts is often inferred from personality classification or algorithmic scoring rather than measured as a dynamic and context-sensitive behavioural phenomenon. This separation restricts the ability to comprehensively understand how MBTI personality dimensions, emotional regulation, and neural engagement interact during formal presentations, thereby reinforcing the need for an integrated analytical framework.

3.4. Anticipated Challenges and Limitations

Several methodological and analytical challenges are anticipated. Although data are collected in controlled presentation environments, variability in baseline neural activity, presentation topics, and timing may influence measured engagement indicators. The dataset comprises high-frequency electroencephalographic (EEG) bandpower signals recorded at sub-second intervals from approximately 100 participants, requiring substantial preprocessing, temporal segmentation, feature extraction, and noise filtering before clustering analysis. The presence of missing values further increases computational complexity.

Interpreting neural bandpower signals presents an additional challenge, as elevated activity within specific frequency ranges may reflect anxiety, cognitive effort, or heightened engagement rather than stress alone. Self-reported MBTI and emotional intelligence measures introduce potential response bias, capturing self-reported orientation rather than directly observable behaviour. Moreover, clustering techniques must be carefully selected and validated to avoid artificial groupings, particularly given the likelihood of overlapping neural response patterns across personality dimensions.

Finally, the observational design permits identification of statistical associations but does not establish causality. Findings may therefore be context-specific to the sampled cohort and institutional presentation setting.

4. AIMS & OBJECTIVES

4.1. Research Aim and Objectives

Research Aim

This study aims to investigate whether neural response patterns recorded during formal presentations align with the Myers–Briggs Type Indicator (MBTI) personality dimensions. By integrating electroencephalographic (EEG) bandpower data from the Muse 2 wearable device with unsupervised clustering analysis and psychometric measures, the study seeks to evaluate whether MBTI personality orientations reliably correspond with measurable neural engagement under performance conditions.

Research Objectives

Objective 1: To preprocess and structure high-frequency EEG bandpower data through data cleaning, participant-level aggregation, feature extraction, and management of missing values in order to establish a reliable analytical dataset.

Objective 2: To apply and validate unsupervised clustering techniques on derived neural features to identify naturally occurring neural engagement patterns during formal presentations without prior personality labelling.

Objective 3: To compare identified neural clusters with each MBTI personality dimension (introversion–extraversion, sensing–intuition, thinking–feeling, judging–perceiving) to assess the extent of alignment or divergence between observed neural engagement patterns and self-reported personality orientations.

Objective 4: To incorporate emotional intelligence measures as an interpretative framework for analysing stress regulation and attentional stability within identified neural clusters.

Objective 5: To design and implement a visual analytical dashboard that presents clustering outcomes and neural engagement patterns in an interpretable, decision-support format.

4.2. Expected Contributions of the Study

This study is expected to contribute in three principal ways.

First, it contributes theoretically by examining whether MBTI personality dimensions reliably correspond with measurable neural engagement patterns during high-pressure presentation environments. By analysing physiological evidence alongside psychometric classifications, the study reframes presentation performance as a measurable physiological engagement process rather than a fixed personality-based trait.

Second, it contributes methodologically by integrating wearable electroencephalographic measurement, unsupervised clustering, and psychometric comparison within a unified analytical framework. This addresses the fragmentation identified in existing research by combining personality theory, neural measurement, and data-driven pattern identification into a single model.

Third, it contributes practically by developing a visual analytics dashboard that presents neural engagement patterns in an interpretable format. This provides potential application in educational

assessment and organisational communication training, enabling more evidence-based evaluation and development practices.

5. RESEARCH METHODOLOGY

5.1 How would you conduct your research?

This study adopts a quantitative, exploratory and data-driven design based on secondary analysis of anonymised electroencephalographic (EEG) data collected during formal presentation sessions. The recordings were obtained from undergraduate students at Middlesex University using the Muse 2 wearable EEG device, which outputs bandpower values across delta, theta, alpha, beta, and gamma frequency ranges.

The EEG dataset consists of time-stamped bandpower measurements recorded at sub-second intervals and exported in Open Sound Control (OSC) format. Each record includes timestamp information, participant identifier, and bandpower values (osc_data_Delta, osc_data_Theta, osc_data_Alpha, osc_data_Beta, osc_data_Gamma). Personality classifications (MBTI) and emotional intelligence (EI) scores are provided in a separate dataset indexed by participant ID. These datasets will be integrated prior to analysis.

5.1.1 Data Preprocessing

The integrated dataset will be cleaned to remove missing values and zero-only bandpower rows. Because recordings are captured at high temporal resolution, data will be organised at the participant level, ensuring each presentation session is analysed independently. Derived features will be standardised to prevent scale differences from influencing clustering outcomes.

5.1.2 Feature Extraction

To reduce high-frequency time-series data into analytically meaningful representations, participant-level features will be computed. These include:

- Mean alpha bandpower
- Mean beta bandpower
- Standard deviation of beta bandpower
- Mean beta-to-alpha ratio

These variables represent overall engagement level, engagement stability, and relative activation. The resulting feature matrix will form the basis of clustering. Data preprocessing and clustering will be implemented in Python, while Tableau will be used for visual analytics development.

5.1.3 Clustering and Validation

Unsupervised clustering (K-means) will be applied to the participant-level feature matrix to identify naturally occurring neural engagement patterns without prior personality labelling. The optimal number of clusters will be determined using the elbow method and silhouette score.

Cluster validity will be further assessed using internal evaluation metrics, including within-cluster sum of squares and silhouette coefficients, to ensure meaningful separation between neural engagement groups. Feature scaling will be verified prior to clustering to ensure comparability across neural engagement variables.

5.1.4 Comparative Statistical Analysis

Identified neural engagement clusters will be compared with MBTI personality dimension orientations and emotional intelligence (EI) scores. Separate chi-square tests will be conducted for each MBTI dimension (introversion–extraversion, sensing–intuition, thinking–feeling, judging–perceiving) to assess whether personality orientations are unevenly distributed across neural clusters.

Analysis of variance (ANOVA) will be used to evaluate whether mean EI scores differ significantly between identified neural engagement groups. Where appropriate, post-hoc comparisons will be conducted to examine specific group differences.

5.1.5 Visual Analytics Development

A series of visual analytics dashboards will be developed using Tableau to present clustering outcomes and neural engagement metrics in an interpretable format. The dashboards will include:

- Participant-level neural engagement summaries
- Cluster distribution visualisations
- Comparative visualisations of MBTI classifications across clusters
- Emotional intelligence score distributions by cluster

Interactive filtering will allow exploration of engagement variability and cluster composition. These dashboards will translate analytical findings into accessible visual insights, supporting evidence-based interpretation of neural engagement patterns in formal presentation settings.

5.1.6 Ethical Considerations

The study involves secondary analysis of anonymised EEG and psychometric data collected from undergraduate students at Middlesex University. No direct personal identifiers are included, and no additional data collection will be conducted.

6. PROJECT TIMELINE (GANTT CHART)

The project will be conducted over a structured 12-week development period.

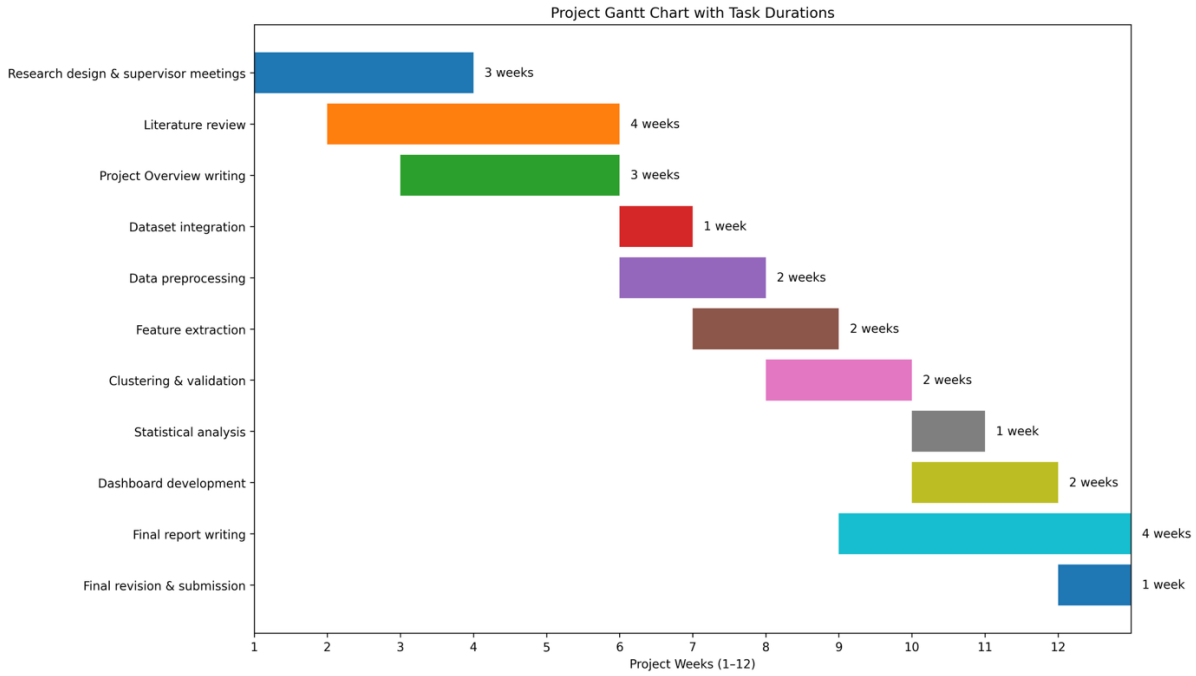

Figure 4. Project Gantt Chart illustrating the timeline and sequencing of research activities (Weeks 1–12).

The project timeline is structured into two main phases. Weeks 1-6 focus on literature review, research design refinement, and completion of the Project Overview. This stage establishes the conceptual framework, confirms the analytical scope, and finalises the methodological approach prior to technical implementation.

Weeks 6-12 focus on analytical development and evaluation. During this phase, the electroencephalographic (EEG) dataset will be integrated, cleaned, and transformed into participant-level feature representations. Unsupervised clustering, statistical analysis, and visual analytics development will be conducted iteratively, with the final weeks dedicated to interpretation of findings, report consolidation, and submission preparation.

7. CONCLUSION

This project proposes a data-driven investigation into whether Myers-Briggs Type Indicator (MBTI) personality dimensions meaningfully correspond with neural engagement patterns during formal presentations. While personality classifications are frequently used to interpret differences in confidence and communication performance, existing evidence suggests that such psychometric tools do not reliably predict observed behaviour under evaluative conditions. This study, therefore, shifts the analytical focus from self-reported personality orientation to measurable physiological indicators.

Using anonymised electroencephalographic (EEG) bandpower data collected from undergraduate students at Middlesex University through the Muse 2 wearable device, the project applies participant-level feature extraction and unsupervised clustering to identify naturally occurring neural engagement patterns. These physiologically derived clusters will then be compared with each MBTI personality dimension (introversion-extraversion, sensing-intuition, thinking-feeling, judging-perceiving) and emotional intelligence scores to examine potential alignment or divergence between personality orientations and observed neural responses.

By integrating neural measurement, data-driven clustering, and psychometric comparison within a unified analytical framework, the study seeks to provide a more empirically grounded understanding of presentation performance. Rather than presuming that personality dimensions determine behaviour in formal presentation contexts, the project evaluates whether neural evidence substantiates, complicates, or challenges personality-based assumptions.

In doing so, it advances a shift from interpretive classification toward measurable engagement, contributing to a more evidence-based foundation for understanding performance under evaluation.

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